Chapter 5. Seasonal Time Series Models

Seasonal components and seasonal ARIMA

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Many series repeat a pattern with known period s ($s = 12$ monthly, $s = 4$ quarterly): *seasonality*. This chapter adapts ARIMA models to seasonal data.

5.1 Time series with a seasonal component

Motivation. A periodic component is not removed by ordinary differencing, but differencing at lag s removes it.

Suppose

$$X_t = S_t + Z_t, \tag{5.1}$$

with *seasonal component* $S_t = S_{t+s}$, $S_1, \ldots, S_s$ independent, $ES_t = 0$, $\mathrm{Var}(S_t) = \sigma_t^2 < \infty$, and $\{Z_t\}$ stationary, mean 0, ACVF γ_Z, independent of $\{S_t\}$. Then

$$\begin{aligned} EX_{t+h}X_t &= E(S_{t+h} + Z_{t+h})(S_t + Z_t) = ES_{t+h}S_t + \gamma_Z(h) \\ &= \begin{cases} \sigma_t^2 + \gamma_Z(h), & h = 0, \pm s, \pm 2s, \ldots, \\ \gamma_Z(h), & \text{otherwise.} \end{cases} \end{aligned}$$

This depends on t, and even with equal σ_t^2 it does not decay along $s, 2s, \ldots$, so no causal ARMA fits $\{X_t\}$.

The *seasonal difference operator*

$$\nabla_s = 1 - B^s$$

gives (*seasonal differencing*)

$$Y_t = (1 - B^s)X_t = X_t - X_{t-s} = S_t + Z_t - (S_{t-s} + Z_{t-s}) = Z_t - Z_{t-s},$$

which is stationary. Fitting $\phi(B)Y_t = \theta(B)\epsilon_t$, $\{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2)$,

$$\phi(B)\nabla_s X_t = \theta(B)\epsilon_t. \tag{5.2}$$

Definition 5.1.1 (Seasonal difference model)

The model (5.2) is called the *seasonal difference model*.

5.2 Time series with trend and seasonality

Motivation. One seasonal difference removes seasonality and lowers the trend degree by one; ordinary differences remove the rest.

Let

$$X_t = T_t + S_t + Z_t,$$

with mutually independent trend, seasonal and stationary parts,

$$T_t = \beta_0 + \beta_1 t + \cdots + \beta_r t^r \quad (r \geq 1,\ \beta_r \neq 0), \qquad S_t = S_{t+s} \text{ for all } t.$$

Then $\nabla_s X_t = \nabla_s T_t + \nabla_s Z_t$ since $\nabla_s S_t = 0$.

Lemma (Removing a polynomial trend)

With T_t and Z_t as above,
(1) $(1-B)^{r-1}\nabla_s T_t = \mu$ for the constant $\mu = s\,r!\,\beta_r$;
(2) $(1-B)^{r-1}\nabla_s Z_t$ is stationary with mean 0.
Consequently

$$(1-B)^{r-1}\nabla_s X_t = (1-B)^{r-1}\nabla_s T_t + (1-B)^{r-1}\nabla_s Z_t = \mu + (1-B)^{r-1}\nabla_s Z_t. \tag{5.3}$$

Proof. [Supplement]

Step 1. For $P(t) = at^k + \cdots$, $k \geq 1$, and $m \geq 1$: $t^k - (t-m)^k = km\,t^{k-1} + \cdots$, so $P(t) - P(t-m)$ has degree $k-1$ and leading coefficient akm (0 if $k = 0$).

Step 2. With $m = s$ once and $m = 1$ another $r-1$ times, the leading coefficient becomes $\beta_r rs \cdot (r-1)! = s\,r!\,\beta_r$ at degree 0: (1).

Step 3. $(1-B)^{r-1}\nabla_s = \sum_{j=0}^{m} c_j B^j$, $m = r-1+s$, so $U_t = \sum_j c_j Z_{t-j}$ has $EU_t = 0$ and $\mathrm{Cov}(U_{t+h}, U_t) = \sum_{j,k} c_j c_k \gamma_Z(h-j+k)$, free of t: (2). Linearity gives (5.3). ■

Fitting ARMA to $(1-B)^{r-1}\nabla_s X_t - \mu$:

$$\phi(B)\{(1-B)^{r-1}\nabla_s X_t - \mu\} = \theta(B)\epsilon_t, \qquad \{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2). \tag{5.4}$$

Definition 5.2.1 (Seasonal difference model with trend)

The model (5.4) is called the *seasonal difference model with trend*.

For $r = 1$ one seasonal difference suffices; in practice ∇_s is often followed by one ∇.

5.3 Seasonal ARIMA models

Motivation. Stochastic seasonality, where an observation depends on the one a season earlier, is modelled by ARMA recursions in B^s.

The model

$$X_t = \phi X_{t-s} + \epsilon_t, \qquad \{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2), \tag{5.5}$$

has, for $|\phi| < 1$, the stationary solution

$$X_t = \sum_{i=0}^{\infty} \phi^i \epsilon_{t-si}. \tag{5.6}$$

It has no component S_t, yet $\gamma_X(h) = 0$ unless $s \mid h$: a seasonal ARMA (SARMA) effect. Allowing seasonal differencing gives SARIMA.

Definition 5.3.1 (SARIMA$(P, D, Q)_s$ model)

A time series $\{X_t\}$ follows a *seasonal ARIMA model* SARIMA$(P, D, Q)_s$ if

$$\Phi(B^s)\{(1-B^s)^D X_t - \mu\} = \Theta(B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2), \tag{5.7}$$

where

$$\Phi(z) = 1 - \Phi_1 z - \cdots - \Phi_P z^P, \qquad \Theta(z) = 1 + \Theta_1 z + \cdots + \Theta_Q z^Q$$

satisfy the causality and invertibility conditions. Usually $\mu = 0$ is assumed.

It suits data correlated only at the same seasonal position; $D \neq 0$ handles a seasonal trend. Models (5.4) and (5.7) differ; the data decide.

Example 5.3.2 (ACF of seasonal MA and AR models)

(1) For MA$(2)_s$, $X_t = \epsilon_t + \theta_1\epsilon_{t-s} + \theta_2\epsilon_{t-2s}$, the MA(2) ACVF sits on lags $0, s, 2s$:

$$\gamma_X(h) = \begin{cases} (1+\theta_1^2+\theta_2^2)\sigma^2, & h = 0, \\ (\theta_1 + \theta_1\theta_2)\sigma^2, & h = s, \\ \theta_2\sigma^2, & h = 2s, \\ 0, & \text{otherwise } (h \geq 0). \end{cases}$$

(2) For AR$(1)_s$, $X_t - \phi X_{t-s} = \epsilon_t$, $|\phi| < 1$, (5.6) gives $\gamma_X(sj) = \sigma^2 \sum_{i\geq 0} \phi^i\phi^{i+j} = \phi^j\sigma^2/(1-\phi^2)$, $\gamma_X(h) = 0$ if $s \nmid h$:

$$\rho(h) = \begin{cases} \phi^{h/s}, & h = 0, s, 2s, \ldots, \\ 0, & \text{otherwise.} \end{cases}$$

5.4 Multiplicative seasonal models

Motivation. Multiplying a seasonal operator in B^s by an ordinary one in B captures both within- and across-season dependence with few parameters.

Definition 5.4.1 (Multiplicative seasonal ARIMA model)

A time series $\{X_t\}$ follows a *multiplicative seasonal* ARIMA$(p,d,q)(P,D,Q)_s$ model if

$$\phi(B)\Phi(B^s)(1-B)^d(1-B^s)^D X_t = \theta(B)\Theta(B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0,\sigma^2),$$

with ϕ, θ as in Definition 4.1.1 and Φ, Θ as in Definition 5.3.1.

It contains Definitions 5.1.1, 5.2.1, 5.3.1; its ACF shows a short-lag pattern (ϕ, θ) repeated around $s, 2s, \ldots$ (Φ, Θ).

Example 5.4.2 (The ARIMA$(0,1,1)(0,1,1)_s$ model)

The model $\nabla_s(1-B)X_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$ reads

$$X_t - X_{t-1} - X_{t-s} + X_{t-s-1} = \epsilon_t + \theta\epsilon_{t-1} + \Theta\epsilon_{t-s} + \theta\Theta\epsilon_{t-s-1}.$$

$W_t = \nabla_s(1-B)X_t$ is the stationary ARIMA$(0,0,1)(0,0,1)_s$. For $s \geq 3$,

$$\gamma_W(0) = (1+\theta^2)(1+\Theta^2)\sigma^2, \qquad \gamma_W(1) = \theta(1+\Theta^2)\sigma^2,$$
$$\gamma_W(s) = \Theta(1+\theta^2)\sigma^2, \qquad \gamma_W(s-1) = \gamma_W(s+1) = \theta\Theta\sigma^2,$$

and $\gamma_W(h) = 0$ for $2 \leq h \leq s-2$ and for $h \geq s+2$ (Figure 1).

Example 5.4.3 (The ARIMA$(0,0,1)(1,0,1)_s$ model)

Let $(1-\Phi B^s)X_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$, $|\Phi| < 1$, $s \geq 4$, and $X_t = (1+\theta B)Y_t$ with

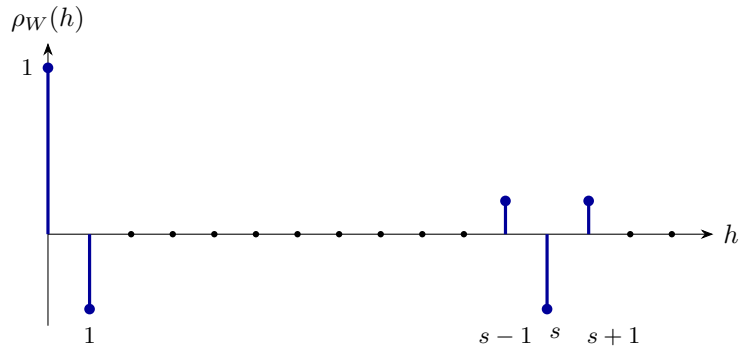


Figure 1: ACF of $W_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$ ($\theta, \Theta < 0$, $s = 12$): nonzero only at lags $0, 1, s-1, s, s+1$.

SARMA$(1,1)_s$ $Y_t = \Phi Y_{t-s} + \epsilon_t + \Theta\epsilon_{t-s}$. Using $\mathrm{Cov}(Y_t, \epsilon_t) = \sigma^2$, $\mathrm{Cov}(Y_t, \epsilon_{t-s}) = (\Phi+\Theta)\sigma^2$,

$$\gamma_Y(0) = \Phi^2\gamma_Y(0) + (1+\Theta^2)\sigma^2 + 2\Phi\Theta\sigma^2 \implies \gamma_Y(0) = \Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$
$$\gamma_Y(s) = \Phi\gamma_Y(0) + \Theta\sigma^2 = \Big\{\Theta + \Phi + \frac{\Phi(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$

$\gamma_Y(ks) = \Phi^{k-1}\gamma_Y(s)$, $k \geq 1$, $\gamma_Y(h) = 0$ if $s \nmid h$. With $\gamma_X(h) = (1+\theta^2)\gamma_Y(h) + \theta\{\gamma_Y(h-1) + \gamma_Y(h+1)\}$,

$$\gamma_X(0) = (1+\theta^2)\Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2, \qquad \gamma_X(1) = \theta\Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$
$$\gamma_X(s) = (1+\theta^2)\gamma_Y(s), \qquad \gamma_X(s-1) = \gamma_X(s+1) = \theta\gamma_Y(s),$$

$\gamma_X(2) = \cdots = \gamma_X(s-2) = 0$, $\gamma_X(h) = \Phi\gamma_X(h-s)$ for $h \geq s+2$: the pattern around s recurs around $2s, 3s, \ldots$, damped by Φ.

5.5 Selected exercises

Motivation. Exercises on ψ-weights and autocovariances of seasonal models, and seasonal versus unrestricted MA.

Exercise 5.2 (ψ-weights of a seasonal model)

Let $(1-\Phi B^s)X_t = (1+\theta B)\epsilon_t$ with $|\Phi| < 1$ and $\{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2)$. Find the coefficients a_j in the representation $X_t = \sum_{j=0}^{\infty} a_j\epsilon_{t-j}$ of the stationary solution.

Solution.

Step 1. $1 - \Phi z^s$ vanishes only on $|z| = |\Phi|^{-1/s} > 1$, so by Proposition 2.4.A the unique stationary solution has $a(z) = (1+\theta z)/(1-\Phi z^s)$.

Step 2. For $|z| < |\Phi|^{-1/s}$,

$$a(z) = \sum_{i=0}^{\infty} \Phi^i z^{si} + \theta\sum_{i=0}^{\infty} \Phi^i z^{si+1}.$$

Step 3: $s \geq 2$. The exponents si, $si'+1$ never coincide, so

$$a_{si} = \Phi^i, \qquad a_{si+1} = \theta\Phi^i \quad (i = 0, 1, 2, \ldots), \qquad a_j = 0 \text{ if } j \bmod s \notin \{0, 1\},$$

i.e. $X_t = \sum_{i\geq 0} \Phi^i(\epsilon_{t-si} + \theta\epsilon_{t-si-1})$.

Step 4: $s = 1$. ARMA(1,1): $a_0 = 1$, $a_j = \Phi^{j-1}(\Phi + \theta)$ for $j \geq 1$.

Step 5: check. $a_j - \Phi a_{j-s} = \mathbb{1}\{j = 0\} + \theta\,\mathbb{1}\{j = 1\}$, the coefficient form of $(1 - \Phi z^s)a(z) = 1 + \theta z$. ■

Exercise 5.3 (MA$(1)_{12}$ versus MA(12))

Can the MA$(1)_{12}$ model $X_t = \epsilon_t + \Theta\epsilon_{t-12}$ be regarded as an MA(12) model? Which of the two formulations is preferable, and why?

Solution.

Step 1: it is an MA(12) model. with $\theta_1 = \cdots = \theta_{11} = 0$, $\theta_{12} = \Theta$, and

$$\gamma(0) = (1 + \Theta^2)\sigma^2, \qquad \gamma(12) = \Theta\sigma^2, \qquad \gamma(h) = 0 \text{ for } h \geq 1,\ h \neq 12,$$

since equal-time noise products occur only for $h \in \{0, \pm 12\}$.

Step 2: the same invertibility condition. $1 + \Theta z^{12} = 0$ iff $|z| = |\Theta|^{-1/12}$, outside the unit circle iff $|\Theta| < 1$, the MA$(1)_{12}$ condition.

Step 3: which to prefer. The unrestricted MA(12) estimates eleven zero coefficients, adding $O(n^{-1/2})$ error to fitted ACVFs and forecasts without improving fit (LR statistic $\approx \chi^2(11)$; AIC charges 22, BIC $11 \log n$). By parsimony MA$(1)_{12}$ is preferable and states the structure directly. Only genuine SACF correlation at lags $1, \ldots, 11$ argues otherwise, and then a multiplicative model (Example 5.4.2) is tried first. ■

Exercise 5.4 (ACF of ARIMA$(1,0,1)(0,0,1)_s$)

The book asks for the autocovariance functions of four seasonal models; parts (1), (2) and (4) are worked out in Examples 5.4.3 (the process Y_t there), 5.4.2 and 5.4.3. Part (3): let $s \geq 2$, $|\phi| < 1$ and

$$(1 - \phi B)X_t = (1 + \theta B)(1 + \Theta B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0, \sigma^2).$$

Find the autocovariance function of the stationary solution.

Solution.

Step 1: reduction to ARMA(1,1). Let $(1-\phi B)Y_t = (1+\theta B)\epsilon_t$ (causal) and $X_t = Y_t + \Theta Y_{t-s}$. Then

$$(1 - \phi B)X_t = (1 - \phi B)Y_t + \Theta(1 - \phi B)Y_{t-s} = (1 + \theta B)\epsilon_t + \Theta(1 + \theta B)\epsilon_{t-s} = (1 + \theta B)(1 + \Theta B^s)\epsilon_t,$$

and X is stationary, the unique stationary solution by Proposition 2.4.A.

Step 2: ACVF of Y. By Example 2.4.3, with $c = \gamma_Y(1)$,

$$\gamma_Y(0) = \frac{\sigma^2(1 + 2\phi\theta + \theta^2)}{1 - \phi^2}, \qquad c = \frac{\sigma^2(\phi + \theta)(1 + \phi\theta)}{1 - \phi^2}, \qquad \gamma_Y(h) = c\,\phi^{|h|-1}\ (h \neq 0).$$

Step 3: general formula. By bilinearity,

$$\begin{aligned}\gamma_X(h) &= \text{Cov}(Y_{t+h} + \Theta Y_{t+h-s},\ Y_t + \Theta Y_{t-s}) \\ &= (1 + \Theta^2)\gamma_Y(h) + \Theta\{\gamma_Y(h - s) + \gamma_Y(h + s)\}.\end{aligned}$$

Step 4: explicit values. For $h \geq 0$:

$$\gamma_X(0) = (1 + \Theta^2)\gamma_Y(0) + 2\Theta c\,\phi^{s-1},$$

$$\gamma_X(h) = c\{(1+\Theta^2)\phi^{h-1} + \Theta(\phi^{s-h-1} + \phi^{s+h-1})\}, \quad 1 \le h \le s-1,$$
$$\gamma_X(s) = (1+\Theta^2)c\,\phi^{s-1} + \Theta\{\gamma_Y(0) + c\,\phi^{2s-1}\},$$
$$\gamma_X(h) = c\{(1+\Theta^2)\phi^{h-1} + \Theta(\phi^{h-s-1} + \phi^{h+s-1})\}, \quad h \ge s+1.$$

So $\gamma_X(h) = \phi\,\gamma_X(h-1)$ for $h \ge s+2$, with a seasonal spike $\Theta\gamma_Y(0)$ at $h = s$; $\rho_X(h) = \gamma_X(h)/\gamma_X(0)$. ■